Assignment: Fundamentals of Artificial Intelligence

Section A: Basic Questions

1. Define Artificial Intelligence. Explain its scope in modern computing.
2. List any four major historical milestones in the evolution of AI.
3. Explain the contribution of **Alan Turing** to the field of Artificial Intelligence.
4. What is a **Turing Machine**? Why is it important in computing theory?
5. What is the **Turing Test**? What does it try to measure?
6. Differentiate between **AI ethics** and **Responsible AI**.
7. Write short notes on the **social, legal, and ethical implications** of AI.
8. What is meant by the **structure of an AI system**? Name its main components.
9. State any four **current trends in Artificial Intelligence**.
10. What is the difference between **structured data** and **unstructured data**? Give one example of each.

Section B: Descriptive & Conceptual Questions for understanding

11. Explain the **AI revolution** and discuss how AI is changing real-world industries such as healthcare, education, agriculture, or finance.
12. Discuss the idea of **"being human in the age of AI."** How does AI affect decision-making, creativity, and employment?
13. Explain **artificial life**, **learning through interaction**, and **emergent behavior** with suitable examples.
14. Compare **rules-based systems** and **expert systems**. Where are they useful?
15. Explain the difference between **supervised learning** and **unsupervised learning** with suitable examples.
16. What is **knowledge representation** in AI? Explain any three techniques used for representing knowledge.
17. Explain **state-space representation** with an example of a problem that can be solved using it.
18. What are **search problems** in AI? Explain the difference between **classical search** and **beyond classical search**.
19. Explain the role of **intelligent agents** in AI. What are the characteristics of an intelligent agent?
20. Write notes on:

    - Propositional logic
    - Predicate logic
    - Logic programming

Section C: Case-study-Based / Analytical Questions/Scenario based Questions

21. A college wants to build an AI-based system to predict whether a student may need academic support based on attendance, marks, and assignment submission history.
    Identify:

    - the type of learning to be used,
    - the input data,
    - the expected output,
    - and one ethical concern in this application.

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22. A shopping website recommends products to customers based on past browsing and purchase behavior.
    Explain whether this is **supervised learning**, **unsupervised learning**, or both. Justify your answer.
23. Consider a **medical diagnosis system** that uses predefined rules from doctors to identify possible diseases.
    Is this a **rules-based system**, **expert system**, or machine learning system? Explain your reasoning.
24. A robot is placed in a maze and must find a path from the start point to the goal point.
    Represent this as a **state-space problem** by identifying:

    - initial state,
    - goal state,
    - possible actions,
    - and path cost.

25. A delivery company wants to assign routes to 100 vehicles in a way that minimizes total fuel cost and delivery time.
    Explain why this is a **combinatorial optimization problem**. Suggest how AI can help solve it.
26. In a chess-playing program, the machine must predict and respond to the opponent's next move.
    What type of search is used here? Why is it called **adversarial search**?
27. A smart home assistant receives voice commands, turns devices on or off, adjusts room temperature, and learns user preferences over time.
    Explain how this system acts as an **intelligent agent**.
28. A company collects customer reviews, emails, invoices, and sales records.
    Classify the data into **structured** and **unstructured** categories. Explain how **Big Data** techniques can help analyze them.
29. A facial recognition system is deployed in a public place.
    Analyze this case from the perspective of:

    - AI ethics,
    - responsible AI,
    - privacy,
    - and legal concerns.

30. Design a simple AI solution for one real-life problem from your surroundings, such as:

    - library book recommendation,
    - crop disease detection,
    - traffic signal control,
    - or attendance monitoring.
      Your answer should include:
    - problem statement,
    - type of AI technique used,
    - data required,
    - possible benefits,
    - and risks or limitations.

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